What Do Provably Powerful GNNs Learn? Interpreting Money-Laundering Detectors on an Expressiveness Ladder

CS224W Project Milestone (Project 1: Application)

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Milestone submission, 2026-11-11. Code: `src/amlinterp/` (data processing, models, training and evaluation, analysis), with unit tests in `tests/`. Every number here is traceable to `docs/experiment_log.md`.

1. Problem and motivation

Money launderers move funds through recognizable transaction patterns: fan-ins, fan-outs, scatter-gathers, and cycles that return money to its origin. Message-passing GNNs are at most as expressive as 1-WL and provably cannot count many such substructures. Egressy et al. (AAAI 2024) showed that three adaptations (reverse message passing, port numbering and ego IDs) make a GNN able to detect any directed subgraph pattern in a directed multigraph, and that they greatly improve laundering detection on IBM's AMLworld benchmark.

Higher F1 does not show how the improved models work, and a compliance analyst needs to know whether a flag reflects a laundering pattern or a shortcut. Our question:

> When an adaptation provably unlocks a pattern, does the trained model develop internal features that detect it, and does it rely on them?

We train an expressiveness ladder of four models that differ by exactly one adaptation each, and test three hypotheses:

- **H1:** concept coverage (how well the best single unit detects a pattern) grows along the ladder as the theory predicts. For example, cycle detectors appear only once ego IDs are added.
- **H2:** the new detectors are causally used, so mean-ablating them hurts that pattern specifically.
- **H3:** where no single neuron detects a provably detectable pattern, sparse-autoencoder latents recover it.

This milestone reports:

- the data pipelines;
- the trained ladder on synthetic data (a reproduction of Egressy et al. Table 1) and on AMLworld (a reproduction of Table 2);
- the first dissection results (H1), including a depth experiment that tests one of its predictions.

2. Datasets and processing

2.1 Synthetic graphs with exact pattern labels

We follow Egressy et al. App. D. Each graph is a random circulant directed multigraph:

- n = 8,192 nodes and m = ⌊nd/2⌋ = 24,576 edges (average degree d = 6);
- each edge picks u uniformly and v = round(N(u, r)) mod n with r = 11.1;
- self-loops are re-drawn;
- a random edge time orders the port numbers.

Train, validation and test are three independently generated graphs.

Every node gets 11 binary labels, computed by enumerating each pattern in the graph:

- deg-in/out > 3 (counting multi-edges);
- fan-in/out > 3 (counting unique neighbors);
- lies on a directed k-cycle, k = 2..6 (C2–C6);
- scatter-gather (S-G) sink;
- biclique (B-C, the sink of a directed K2,2).

The labelers are unit-tested against brute force on small graphs.

Change from the proposal. The proposal said patterns would be planted. We instead label the patterns that occur naturally in random graphs, as Egressy et al. do. Planting adds extra edges, so pattern nodes would have higher degree than other nodes. A degree neuron would then look like a cycle detector. This "degree shortcut" is exactly the confound an interpretability study has to avoid.

Re-drawing self-loops matters: keeping them undershoots Egressy's positive rates by 2–5 points on C4–C6, S-G and B-C. With re-drawing, the train-graph rates match their Table 3 within about one sd (deg-in .348, C2 .200, C6 .784, S-G .322, B-C .322). The base rates range from 0.19 (C2) to 0.78 (C6). That range is why we use rate-matched thresholds and chance-corrected scores (Section 4).

2.2 AMLworld HI-Small

AMLworld (Altman et al., NeurIPS 2023) is a synthetic transaction dataset from an agent-based simulator. We use HI-Small:

- 515,088 accounts (nodes) and 5,078,345 transactions (directed, timestamped edges with amount, currency and payment format);
- 5,177 laundering transactions (0.10%).

The task is binary edge classification. We follow IBM's Multi-GNN preprocessing exactly, so our results are comparable to Egressy Table 2:

- **Chronological day split.** It comes out at 64/19/17%, not the nominal 60/20/20. The illicit ratio rises across splits (0.078% → 0.107% → 0.186%), so thresholds tuned on validation see a shifted base rate on test.
- **Nested split graphs.** Train ⊂ train+val ⊂ all; each split is evaluated only on its own edges.
- **Per-split z-normalization** of edge features.
- **Ports.** Each account numbers its counterparties in order of their first transaction. All edges with the same counterparty share a port, and reverse edges swap the in- and out-port columns. IBM computes ports with a Python loop over 5M edges. We vectorized it (tested equal to IBM's loop) and cache the result.

`HI-Small_Patterns.txt` lists the transactions of 370 laundering attempts of 8 pattern types:

- 3,209 rows, all of which we match to edges;
- they cover 62% of laundering transactions;
- 147 attempts span a split boundary.

These pattern labels are the ground-truth concepts for the AML dissection. Two properties of the graph matter for interpretation:

- 11.6% of edges are self-loops and 80% are parallel edges;
- with the benchmark depth of 2 layers, only cycles of length 2–3 through the target edge are detectable, and the test split has just 6 such cycle attempts.

3. Models

All rungs share the GIN-with-edge-features backbone of IBM Multi-GNN (PyG):

- linear node and edge embeddings;

- residual layers x ← (x + ReLU(BN(GINEConv(x, e)))) / 2;
- hidden size 64;
- the readout MLP(→50→25→2) on [h_u ‖ h_v ‖ e_uv] for edges (AML) or on h_v for nodes (synthetic, 11 outputs).

Rung	Adds	Theory: newly detectable
R0	GIN	in-degree, edge statistics
R1	+ reverse message passing (separate `to`/`rev_to` convolutions, averaged)	out-degree, fan-out
R2	+ port numbers (in- and out-port as edge features)	fan-in, fan-out
R3	+ ego IDs (a 0/1 feature on the target's endpoints) = Multi-GIN	cycles, scatter-gather, bicliques

Controlled experiment. Every rung uses the same hyperparameters, split, epoch budget, model-selection rule (best validation mean F1) and seeds (1, 2, 3). Decision thresholds and checkpoints are chosen on validation only.

- **Synthetic:** 6 layers (Egressy F.3, so 6-cycles are in range), full neighborhoods, Adam with lr 3e-3, 100 epochs, batch 256. The lr is the best of {3e-4, 1e-3, 3e-3} by validation F1 averaged over rungs.
- **AML:** IBM's GIN settings (lr 6.2e-3, class-weighted cross-entropy, [100, 100] neighbor sampling, batch 8,192, 2 layers, 100 epochs).

Ego IDs. IBM's code marks the endpoints of all seed edges in a batch, not just the scored edge. We implement both versions:

- `batch` reproduces the benchmark and is used for AML training;
- `exact` marks only the target and is used for synthetic training and for all synthetic activations.

R3's activations depend on this choice, so it is recorded with every saved activation file.

4. Interpretability methods and metrics

Task metrics. Minority-class F1 (the benchmark standard) at the 0.5 threshold and at a validation-tuned threshold, plus PR-AUC.

Activations. For each trained model we record, for every test item, all hidden layers h0..h6 and both decoder layers. On synthetic data an item is a node, giving 523 units. On AML an item is an edge, and the representation is [h_u ‖ h_v] plus the decoder layers.

Neuron dissection (H1). For unit k and binary concept c we compute two scores:

- **IoU** = |M_k ∩ C_c| / |M_k ∪ C_c|, where the unit's mask M_k is its top-p_c fraction of items. Matching the concept's own base rate p_c is needed because the concepts are common (0.19–0.78) and a fixed top quantile would cap IoU.
- **Average precision (AP)** of the unit's activation as a score for the concept. It is threshold-free, and chance is p_c. We report it chance-corrected, (AP − p_c)/(1 − p_c), so 0 means a random unit and 1 a perfect detector.
- **AUROC**: the probability that a random concept item activates the unit more than a random non-concept item. Ties count one half, since tied items are ones the unit cannot tell apart; in a GNN these are often structurally identical nodes. Chance is 0.5 for every concept, so AUROC is the headline metric on synthetic data. AP is the headline for the rare AML concepts, where AUROC is dominated by easy negatives. AP is the same estimator as the models' PR-AUC, so a unit can be compared directly with the full model's output.

Hidden states can be negative, so each unit is scored in both directions. Concept coverage is the best score over units, overall and per layer.

The best of hundreds of units beats single-unit chance easily, so we test it against a **permutation null**: concept labels are shuffled across items 20 times and the max over all units recomputed. A **bootstrap** (100 resamples, re-selecting the best unit each time) gives CIs. On AML, resampling is at the level of laundering attempts.

For the precision-recall curves, each concept's unit is selected on the validation graph and evaluated on the test graph. Picking the best of hundreds of units on the same data it is scored on would inflate the curves.

Coverage is reported over the GNN layers h1..h6. The decoder layers sit directly before the output, so any accurate model "covers" its own targets there.

Planned for H2/H3 (after the milestone):

- mean-ablation of detector units, with training-split means and validation thresholds fixed beforehand, measuring per-concept recall drops against random-unit ablations;
- TopK sparse autoencoders (Gao et al. 2024) on the hidden states, followed by the same dissection on their latents;
- linear probes as a decodability ceiling.

5. Preliminary results

5.1 The synthetic ladder reproduces Egressy et al. Table 1

Test minority-class F1 (%), mean over 3 seeds at the 0.5 threshold; sd in parentheses where it is ≥ 2 points (and for the row means and R3 C6). Egressy Table 1 (mean of 5 runs) in brackets.

Rung	deg-in	deg-out	fan-in	fan-out	C2	C3
R0 GIN	99.9 [99.8]	20.9 (10.8) [43.6]	95.3 [95.6]	14.9 (9.3) [35.9]	27.0 (6.3) [34.7]	53.3 (4.6) [58.0]
R1 +reverse	99.3 [98.9]	99.3 [99.1]	95.2 [95.0]	96.0 [95.3]	49.0 (2.2) [36.0]	64.4 (2.0) [63.9]
R2 +ports	99.8 [98.4]	99.2 [98.4]	99.8 [98.5]	99.7 [99.2]	47.9 [39.2]	63.5 [63.6]
R3 +ego IDs	99.2 [99.5]	99.8 [99.1]	100.0 [99.6]	99.9 [99.3]	100.0 [99.0]	100.0 [98.7]

Rung	C4	C5	C6	S-G	B-C	mean
R0 GIN	64.3 (3.3) [50.8]	52.7 [43.1]	52.3 [48.6]	68.3 [69.3]	64.9 (2.2) [63.1]	55.8 (2.6) [58.4]
R1 +reverse	74.1 [69.1]	73.6 [67.4]	77.6 [71.2]	68.7 [65.8]	67.5 (2.2) [66.2]	78.6 (0.4) [75.3]
R2 +ports	75.7 [69.0]	73.4 [70.4]	76.5 [75.0]	67.5 [67.8]	66.3 [65.8]	79.0 (0.2) [76.8]
R3 +ego IDs	99.7 [97.5]	98.2 [91.6]	78.3 (0.6) [84.2]	98.9 [97.4]	98.3 [94.3]	97.5 (0.1) [96.4]

Each adaptation unlocks what the theory predicts:

- **R1 (reverse):** out-degree jumps from 21 to 99 and fan-out from 15 to 96.
- **R2 (ports):** fan-in and fan-out reach ≥ 99.7.
- **R3 (ego IDs):** C2–C5, scatter-gather and biclique jump to 98–100. On average the ladder matches or slightly exceeds the paper's.

Deviations, logged as found:

- **R0 out-degree and fan-out** are below the paper (20.9 vs 43.6, 14.9 vs 35.9) with high variance across seeds. A plain GIN cannot see outgoing edges, so these scores are near chance by construction. The paper's higher values likely reflect threshold or seed noise more than real signal.
- **R1/R2 2-cycles** are about 10 points above the paper.
- **R3 6-cycles** are 6 points below (78.3 vs 84.2).
- **Training length:** 8 of 12 runs selected a checkpoint from epoch ≥ 87, so 100 epochs is slightly short. This applies equally to all rungs, so the comparison stays controlled.

5.2 Dissection: which concepts get a dedicated unit, and where

Setup.

- All 12 synthetic models are dissected on the test graph, with exact ego IDs.
- Each model has 523 units: h0–h6 (64 each) plus the two decoder layers (50 and 25).
- Each unit is scored in both directions against each of the 11 labels.
- A concept's coverage is the score of the best single unit in the GNN layers h1–h6. The decoder is excluded, because it sits next to the output.
- Because that is a best-of-384 choice, its chance level comes from a permutation null: the 95th percentile is about 0.52–0.53 AUROC. Every coverage value below exceeds it in every seed.
- Cross-check: activations re-extracted on a second cluster reproduce the dissection exactly (difference 0.00, same best unit for all 11 concepts).

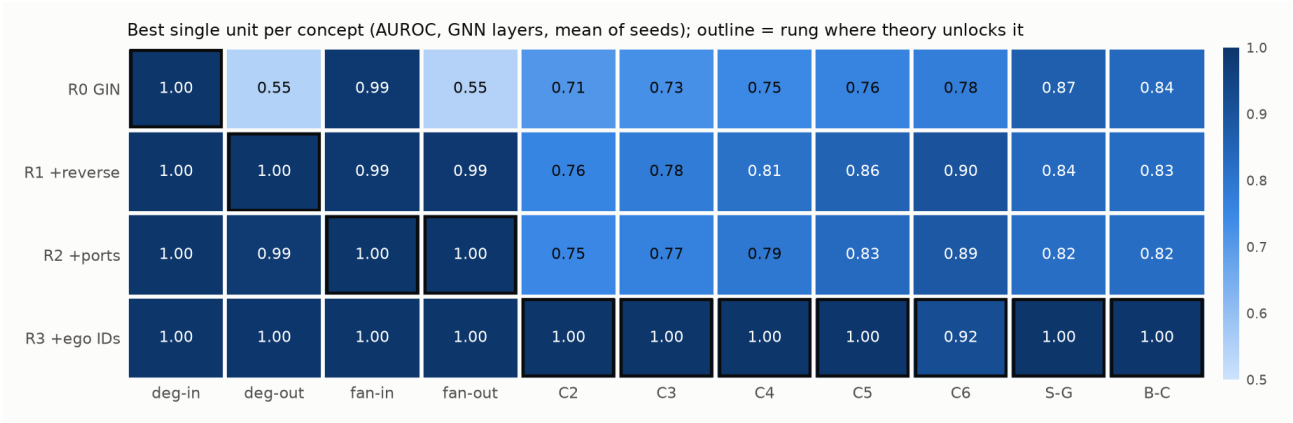


Figure 1: Concept coverage: AUROC of the best single GNN unit (h1–h6) per rung and concept, mean of 3 seeds. Outlined cells mark the rung at which the theory makes the concept detectable.

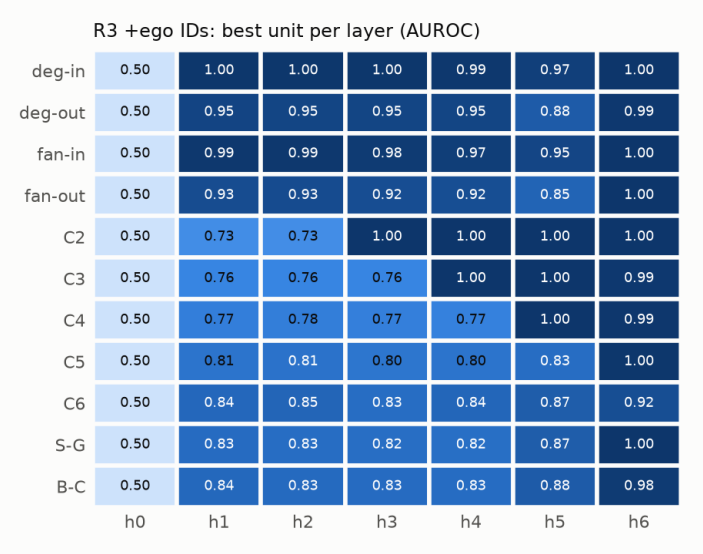


Figure 2: R3 (6 layers): AUROC of the best unit in each layer, per concept, mean of 3 seeds. The k-cycle detector first appears at layer k + 1.

Caveat. On synthetic data the concepts are also the training targets. An accurate model must encode them somewhere, so coverage partly restates Table 1. The two controls below carry the interpretation.

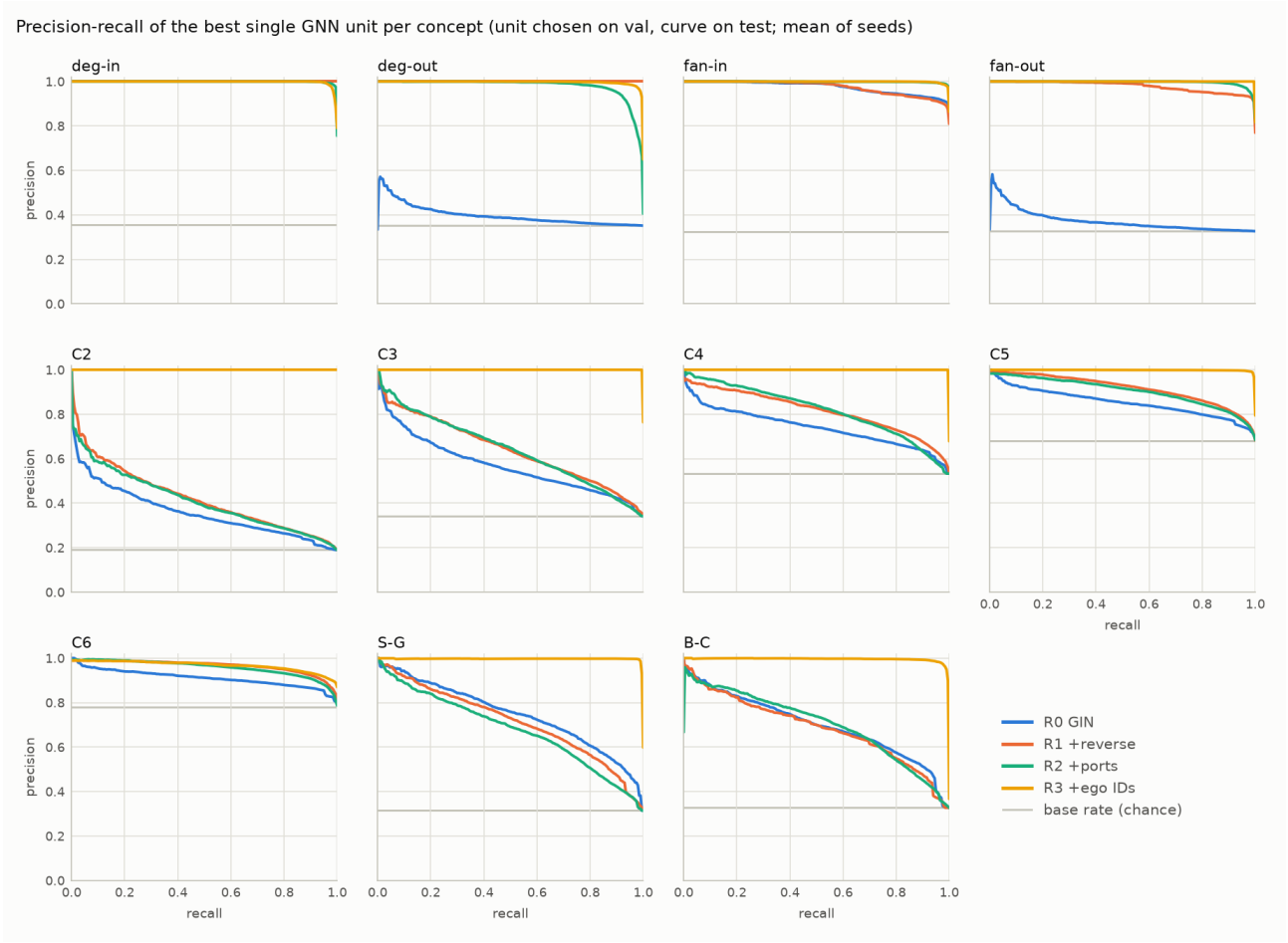


Figure 3: Precision-recall curves of each rung's best GNN unit per concept. The unit is chosen on the validation graph and the curve drawn on the test graph (mean of 3 seeds). The grey line is the concept's base rate.

Finding 1: dedicated detectors appear at the rung the theory predicts, once degree is controlled for. Cycle membership, scatter-gather and biclique sinks all correlate with degree, even in random graphs. Table: AUROC coverage (mean of 3 seeds) next to the best single degree statistic on the same test graph (`degree_baseline.csv`, `amlinterp.analysis.baselines`).

Concept	Degree only	R0	R1	R2	R3
C2	0.74 (total)	0.71	0.76	0.75	**1.00**
C3	0.77 (total)	0.73	0.78	0.77	**1.00**
C4	0.78 (total)	0.75	0.81	0.79	**1.00**
C5	0.81 (total)	0.76	0.86	0.83	**1.00**
C6	0.84 (total)	0.78	0.90	0.89	**0.92**
S-G	0.84 (fan-in)	0.87	0.84	0.82	**1.00**
B-C	0.85 (fan-in)	0.84	0.83	0.82	**1.00**

- **Below R3, the "cycle units" are degree units.** R0–R2 stay within about 0.06 of the degree-only score (C6 under R1: +0.07). R0 sits below total degree because it only sees in-degree (in-degree alone scores 0.68–0.73).
- **At R3, coverage jumps** 0.15–0.26 above the degree baseline for C2–C5, S-G and B-C. This is the jump H1 predicts.
- **Out-degree and fan-out** are absent at R0 (0.55, against an in-degree-only baseline of 0.50) and near-perfect from R1 on.
- **Fan-in** is already covered at R0 (0.99), because parallel edges are rare in these graphs and fan-in is nearly in-degree (in-degree alone: 0.989). The theory's ports step for fan-in can therefore not show up on this dataset.

Finding 2: a k-cycle detector never appears before layer k + 1, and depth is what limits 6-cycles. In the 6-layer R3 models, the first layer with a unit at AUROC ≥ 0.99 is h3 for C2, h4 for C3, h5 for C4 and h6 for C5. This holds in all three seeds (Fig. 2). C6 never gets there (0.90–0.94 at h6).

- The theory only needs k layers for the ego bit to travel around a k-cycle. The trained models need at least one more.
- **Test of the prediction.** We trained R3 with **7 layers**, changing nothing else (seeds 1–3; results in `runs/analysis/depth_R3.md`).
  - C6 test F1 rises from **78.3 ± 0.6 to 93.4 ± 3.3**, above the paper's 84.2.
  - The best C6 unit at h7 reaches AUROC 0.99 / 0.94 / 0.99 (seed 1 crosses 0.99; at 6 layers: 0.93 / 0.94 / 0.90).
  - All other concepts are unchanged (within 0.7 F1).
- **The exact layer is a lower bound, not a fixed rule.** With 7 layers, C2 and C3 stay at h3 and h4, but C4 moves to h6 in two seeds and C5 to h7. Across all six R3 models, no C_k detector appears before h(k+1).
- **So the C6 shortfall at 6 layers is a depth limit, not a training failure.** One extra layer recovers it.

Finding 3: single units suffice at R3. For 10 of 11 concepts (all except C6), an R3 model has one GNN unit with AUROC ≥ 0.99. The PR curves confirm this is not an artifact of picking the best unit on the scored data: units chosen on validation keep precision ≈ 1 up to recall ≈ 1 on test for C2–C5, S-G and B-C (Fig. 3).

Next: concepts the models were not trained on (e.g. cycle counts, other degree thresholds), and linear probes, to separate "represented" from "dedicated unit".

5.3 AMLworld ladder

Three seeds per rung. Regenerate with `python -m amlinterp.analysis.aml_table`, which writes `runs/analysis/aml_ladder_table.{md,csv}`.

Rung	F1 @ 0.5	Egressy T. 2	F1 @ tuned thr.	Thresholds	PR-AUC	Random
R0 GIN	35.23 ± 7.89	28.70 ± 1.13	35.99 ± 8.13	0.60 / 0.55 / 0.60	27.86 ± 9.46	0.19
R1 +reverse MP	53.04 ± 4.74	46.79 ± 4.97	54.03 ± 4.74	0.55 / 0.55 / 0.55	52.92 ± 6.57	0.19
R2 +ports	60.47 ± 3.63	56.85 ± 2.64	61.95 ± 2.40	0.60 / 0.60 / 0.55	60.13 ± 0.99	0.19
R3 +ego IDs	55.66 ± 9.30	57.15 ± 4.99	61.07 ± 3.67	0.75 / 0.50 / 0.75	58.51 ± 3.83	0.19

How to read the columns (all values are percentages; mean ± sd over seeds):

- **F1 @ 0.5** is laundering-class F1 on test when a transaction is flagged at P(laundering) > 0.5. This is IBM's argmax, the number Egressy Table 2 reports.
- **F1 @ tuned thr.** uses the cut-off that maximizes validation F1 (grid 0.05–0.95). Test never influences the choice. **Thresholds** lists the cut-off each seed picked.
- **PR-AUC** (test average precision) is threshold-free. **Random** is a random scorer's expected value: the positive rate, i.e. the test illicit ratio of 0.19%.

Observations:

- **The ladder reproduces Egressy Table 2 within noise.** R1–R3 are within about 1.4 of the paper's sd. Reverse MP is the large step (+17.8 F1; paper +18.1), and ports add another +7.4 (paper +10.1).
- **R0 varies a lot from run to run** (seeds: 38.9 / 40.7 / 26.2). A replicate of seed 1 on a second cluster gave 31.8. R0's mean is above the paper, but the paper's GIN row is copied from Altman et al. (2023), who used different hyperparameters and 4 seeds.
- **R3 vs R2 is unresolved.** At 0.5, R3 has the largest spread (46.7 / 65.3 / 55.0). With validation-tuned thresholds, its spread shrinks to ± 3.7, and R2 and R3 are tied (62.0 vs 61.1). Two of three R3 seeds pick a threshold of 0.75. Much of R3's variance at 0.5 is therefore a calibration shift on test (test has 1.7× the illicit ratio of validation), not worse ranking. The PR-AUCs, 60.1 and 58.5, are also within one sd.
- **The adaptations matter less for AML than for synthetic data.** With 2 layers, ego IDs can only close 2–3-hop cycles. The test split has just 6 such cycle attempts.
- **Hardware:** seed 1 ran on L40S. Seeds 2–3 ran on a mix of L40S, RTX 3090 and H100, recorded per run in `config.json`. GPU nondeterminism is one source of the run-to-run spread.

6. Plan for the rest of the quarter

- **Already in progress.** The mean-ablation pipeline for H2 is implemented and tested (training-split means, validation-selected units and thresholds, size-matched random-unit controls, seed-level t-tests with Benjamini–Hochberg FDR). The synthetic ladder has been retrained with seeds 4–10, so these tests have 10 seeds per rung. Results will be reported in the final report.
- **Nov 11–20:** AML dissection with the confound controls below, and the exact-vs-batch ego comparison.
  - Pattern masks are subsets of the laundering label, so every result is reported against a laundering-label baseline.
  - Structural concepts are computed for all edges, not just laundering ones.
  - CIs are bootstrapped at the attempt level.
- **Nov 20–28:** sparse autoencoders and linear probes (H3) on both datasets, plus concepts the models were not trained on (e.g. cycle counts), to separate "represented" from "dedicated unit".
- **Dec 1–10:** blog post and Colab. The optional Ethereum check is the first item to cut.

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